Zyvor Platform

Complete per-product deep dive — ssahani portfolio

hypercluster v0.1.0 · zyvor.dev · ssahani portfolio

Detailed architecture reference

Companion to PLATFORM_ARCHITECTURE.md — one page per product with stack, capabilities, and integration notes.

19	821	21	14
products documented	lines in ARCH doc	printable PDFs	product deep dives

Section	Deck PDF
Executive summary	06-platform-overview.pdf
Layer model	07-platform-layer-model.pdf
Foundation	08-platform-foundation.pdf
Cloud & apps	09-platform-cloud-apps.pdf
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Per-product detail	14-platform-architecture-detail.pdf
Full reference	PLATFORM-ARCHITECTURE.pdf

hypercluster

Day 0 Kubernetes bootstrap — Kubespray + k3s over SSH

- Single cluster.conf — deploy, scale, upgrade, destroy, kubeconfig
- k3s path: Cilium → cert-manager → MetalLB → KubeVirt/CDI
- forge-handoff --json for Day 2 Forge federation import
- Electron Desktop — Cluster Studio + Liquid Glass UI

Stack & entry points

- Entry: hypercluster bash CLI, lib/k3s.sh, lib/kubespray helpers
- Commands: preflight, deploy, test, health --json, merge-kubeconfig
- Desktop: desktop/ Electron app spawns CLI for lifecycle ops

machina

Enterprise libvirt/KVM hypervisor management — physical layer

- machina-daemon HTTPS :5092 — PAM auth, RBAC, console proxies
- Web UI 40+ pages — noVNC, SPICE, serial, SSH consoles
- Fleet controller + per-host agent for multi-node libvirt
- Prometheus endpoint, alerts, webhooks, backup/restore

Stack & entry points

- Rust workspace: daemon/, core/, web/, tui/, controller/, agent/
- CLI: machinactl deploy · verify · health · backup
- Integrations: OpenStack, Guacamole, KubeVirt migration path

zyvor-fabric

systemd-native private cloud — alternative to libvirt stack

- vmspawnd daemon :8443 — 480+ REST endpoints
- vmctl apply declarative YAML VM specs
- 5 interfaces: CLI, TUI, Web, K8s Operator, Terraform
- GPU passthrough, HA, replication modules

Stack & entry points

- backend/vmspawnd/, vmspawn-driver/, machinectl-driver/
- Runtime: systemd-vmspawn + systemd-machined + KVM
- integrations/machina/ cross-product desktop client

IronWolf

Metal3 bare-metal automation — 24 CRDs (bmo.metal3.io)

- ClusterClaim hardware allocation — binpack/spread/topology
- FirmwareProfile + HardwareBackup fleet lifecycle
- BMC Reconciler — IPMI/Redfish/iLO/iDRAC
- Dashboard :8888 — real-time BMC + host events

Stack & entry points

- Go controller-runtime, config/crd/, web/dashboard/
- terraform-provider-ironwolf/, scripts/quickstart.sh
- Feeds bare metal into hypercluster and Aether Metal3 runtime

v9s · Zeus OS

Visual operating system for KubeVirt and enterprise K8s

- Web dashboard :5050 + TUI with 48+ specialized views
- VM lifecycle, hot-plug, live migration, cost visibility
- OIDC/LDAP/SAML, tenants, audit, scoped API tokens
- Topology/RCA hooks, Velero integration

Stack & entry points

- Rust workspace: core/, tui/, ui/, cli/, chart/
- kube-rs → KubeVirt VirtualMachine/VMI CRs
- Hosts Hermes application layer on Zeus OS clusters

VMRogue · Veyron

KubeVirt command center — GitOps CRDs and Mission Control

- VeyronVM + VeyronPolicy CEL deny rules
- 44 OS templates, 8 resource profiles, blueprint engine
- Direct K8s VNC WebSocket — no virtctl timeout
- GuestKit pre-flight + PacketWolf network context

Stack & entry points

- Rust veyron binary :8080, Go operator/, charts/veyron/
- src/kube/guestkit_client.rs, src/gitops/, terraform-provider-vmrogue/

ragnarok

AI-powered VM management — NL provisioning and cost optimizer

- Natural language VM creation to KubeVirt specs
- AI healer — autonomous incident detection/remediation
- PostgreSQL-backed ops, Playwright E2E
- Optional Aether attestation via RAGNAROK_URL

Stack & entry points

- backend/ Rust Axum, frontend/ React, operator/, charts/ragnarok/
- docs/AI_NL.md, docs/ARCHITECTURE.md

hermes

Application Operating Layer — Launchpad for Kubernetes apps

- Auto-discover Services cluster-wide + EndpointSlice health
- Gateway /a/{namespace}/{slug}/* reverse proxy + WebSocket
- Recognition: Grafana, Argo CD, Harbor, Zeus, Forge
- Favorites, recents, spotlight search — SQLite catalog

Stack & entry points

- Go controller cmd/hermes-controller/, Rust hermes-server :31847
- charts/hermes/ Helm, ui/ React Launchpad

hypersdk

Multi-cloud VM export — 10+ hypervisor providers

- hypervisord daemon :5080 — 238 REST routes, job queue
- Provider adapters: vSphere, Nutanix, AWS, Azure, GCP, Hyper-V
- Embedded React dashboard with live WebSocket progress
- CE vs Enterprise feature split

Stack & entry points

- cmd/hypervisord/, providers/, daemon/jobs/manager.go
- Upstream of hyper2kvm export → convert/deploy pipeline

hyper2kvm

Enterprise VM migration — any hypervisor to KVM

- VMCraft Python engine — 5–7× faster than shell pipelines
- Offline fixers — 35+ OS versions, Windows VirtIO injection
- h2kweb Go API :5070 — 110+ endpoints, VNC console
- Libvirt → KubeVirt live migration, K8s operator

Stack & entry points

- hyper2kvm/vmcraft/, fixers/, web/h2kweb/, operator/
- Pairs with GuestKit for pre-flight assurance

guestkit

Offline VM disk intelligence — zero boot, zero agents

- Pure Rust — QCOW2, VMDK, RAW without libguestfs
- doctor boot scoring · migrate-plan YAML · fleet analyze
- PyO3 bindings hypersdk-guestkit on PyPI
- Windows registry + MBR/GPT + ext4/NTFS/XFS detection

Stack & entry points

- src/disk/, src/core/, k8s/ KubeVirt hooks
- Used by Veyron src/kube/guestkit_client.rs

Aether

Universal runtime portability — 16 migration paths

- One YAML spec → Podman, K8s, KubeVirt, Metal3
- Intent engine scores cost, latency, reliability
- Blue-green, rolling migration with rollback
- Drift detection + auto-reconciliation loop

Stack & entry points

- src/adapters/, src/migration/, src/drift/, web/ React UI
- Helm, deb/rpm, aether serve :8080

forge

Kubernetes-native AI infrastructure — 60+ CRDs

- API Gateway :24631 · Zeus AI ops :8091 · Intelligence :8090
- GPU operator, AI operator, network intelligence, virtualization
- 24 eBPF programs, 94-page React web-ui, Rust CLI
- Closed-loop FinOps, placement, FabricAutoPolicy gates

Stack & entry points

- services/, operators/, crds/, ebpf/, helm/forge-core/
- Consumes PacketWolf; optional hypercluster handoff import

packetwolf

Kernel-native network intelligence — requires Cilium

- netpred explain — cross-layer root cause analysis
- AutoPolicy: traffic learning → Cilium NetworkPolicy
- Autonomous Healer + process-to-socket attribution
- 71K LOC Rust, 130+ REST routes, 18-tab TUI :8443

Stack & entry points

- src/modules/, netpred-ebpf/, web-api/, chart/
- Intelligence plane for Forge and Veyron handlers

GitHub repository map

Local folder	GitHub repo
Aether	ssahani/Aether
forge	ssahani/forge
guestkit	hypersdk/guestkit
hermes	ssahani/hermes
hyper2kvm-	ssahani/hyper2kvm-
hypercluster	ssahani/hypercluster
hypersdk-	ssahani/hypersdk-
IronWolf	ssahani/IronWolf
machina	ssahani/machina
packetwolf	ssahani/packetwolf
ragnarok	ssahani/ragnarok
v9s	ssahani/zeus-os
VMRogue	ssahani/Veyron
zyvor-fabric	ssahani/zyvor-fabric

Questions or demo?

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